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Ch-04 Analysis of Categorical data

Attributes: A qualitative characteristic/property such as skin colour, gender, employment, deafness etc are termed as 'attributes'.

↓

2 class (formed by presence or absence of attribute)

In this chapter

→ we study association of attributes to find out whether a given attributes shows the relationship with other.

let

Weather	Traffic
Raining	mostly heavy
Sunny	mostly light

we say there is association between weather and traffic.

→ later on position

Notation and terminology

presence of an attribute = A, B, C etc

A = literacy

Absence of an attribute = α , β , γ etc

α = illiteracy

combination of attributes: ~~(AB)~~ AB AB

For example if A represents male and

if B = male and C = female

as A = literacy and α = illiteracy

~~(BA)~~

BA = male illiterate

• C α = female illiterate

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* Class frequency: Number of observations (or counts) that fall into that specific class.

→ If A represents the attribute 'literate', then (A) is the frequency of literate people.

→ If α = illiterate, then (α) = frequency of illiterate people.

freq of positive events = $(A), (AB), (ABC)$ ← class frequencies
freq of negative events = $(\alpha), (\alpha\beta), (\alpha\beta\gamma)$ ←
freq of Mixed events = $(\alpha B), (ABC)$ etc
freq of complementary events = $(\alpha A), (\alpha\beta), (\gamma C)$.

* Order of class:

The order tells us how many attributes are being combined to define that class.

1st order = A, B, α, β → Single attribute

2nd order: These describe the combⁿ of two attributes at once. = $AB, \alpha\beta, A\alpha, B\beta, \dots$

Similarly

3rd order = $ABC, \alpha\beta\gamma, \dots$

0th order = This is the total number of all frequencies (N) without any specifications.

For n -attributes, no. of class = 3^n

$n=1$, no. of class = 2, A, α, N

$n=2$, no. of class = 9, $N, A, \alpha, B, \beta, AB, \alpha\beta, \alpha B, \alpha\beta\gamma$

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Table

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Tu

A

α

Ultimate class frequency :

The classes of the highest order are called the ultimate classes. The frequency of any ultimate class is called ultimate class frequency.

For any two attributes A and B

Highest order class is $n=2$

The ultimate class frequency are
(AB), (A β), (α B) and ($\alpha\beta$)

$$\left\{ \begin{array}{l} 3^2 = 9 \\ N, A, B, \alpha, \\ \beta, AB, A\beta \\ \alpha B, \alpha\beta \end{array} \right.$$

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For two attributes A and B:

	A	α	Total
B	(AB)	(αB)	(B)
β	(A β)	($\alpha\beta$)	(β)
Total	(A)	(α)	N

From table.

$$(A) = (AB) + (A\beta)$$

$$(\alpha) = (\alpha B) + (\alpha\beta)$$

$$(B) = (AB) + (\alpha B)$$

$$(\beta) = (A\beta) + (\alpha\beta)$$

$$N = (A) + (\alpha)$$

$$N = (B) + (\beta)$$

$$= (AB) + (A\beta) + (\alpha B) + (\alpha\beta) = (AB) + (\alpha B) + (A\beta) + (\alpha\beta)$$

Conditions of consistency $\checkmark$ simply means that the data make sense in the real world

1) For one attribute, $(A) \geq 0$, $(\alpha) \geq 0$.

$$N = (A) + (\alpha)$$

5 people in survey don't make any sense.

11) For two attributes

$$(AB) \geq 0$$

$$(A\beta) \geq 0$$

$$(\alpha B) \geq 0$$

$$(\alpha\beta) \geq 0$$

ultimate class frequency.

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2071/2018 What is meant by association of attributes:

Given $N = 820$, $(A) = 250$, $(\alpha B) = 50$ and $(AB) = 35$. Test whether the data is consistent.

→ In statistics, an association of attributes refers to the relationship between two or more qualitative characteristics (attributes) in a population.

When we say two attributes are associated, we mean they are not independent. The presence of one attribute changes the likelihood of the other appearing.

Solⁿ Given

	(AB)	(αB)	Total
B	$(AB) = 35$	$(\alpha B) = 50$	(B)
β	$(A\beta)$	$(\alpha\beta)$	(β)
Total	$(A) = 250$	$(\alpha) =$	$N = 820$

$$(A) = (AB) + (A\beta)$$

$$N = (A) + (\alpha)$$

$$(A\beta) = (A) - (AB)$$

$$(\alpha) = N - (A)$$

$$= 250 - 35$$

$$= 820 - 250$$

$$= 215$$

$$= 570$$

Now Since

$$(\alpha) = (\alpha B) + (\alpha\beta)$$

$$(AB) = 35 \geq 0$$

$$(\alpha\beta) = (\alpha) - (\alpha B)$$

$$(A\beta) = 215 \geq 0$$

$$= 570 - 50$$

$$(\alpha B) = 50 \geq 0$$

$$= 520$$

$$(\alpha\beta) = 520 \geq 0, \text{ the data are consistent.}$$

2078 Practice $N = 1000$, $(A) = 450$, $(B) = 650$, $(AB) = 310$
 $N = 820$, $(A) = 520$, $(\alpha B) = 50$ and $(AB) = 35$.
 Test whether the data is consistent or not.

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